The CCD sensor was invented in 1970 at Bell Labs

Sensory Activity

All you wanted to know about image sensor technology – where it's at, what innovations it's going through, and what the future holds...

Mihir Patkar

It seems that in the end, all it takes to capture the male attention is the ability to let the guy boast: "Mine's bigger". These days, any advertisement for a new phone or camera hypes up the megapixel count. The Mobile World Congress was awash with new handsets touting 8-megapixel cameras, while Samsung is already launching a couple of those on our shores here.

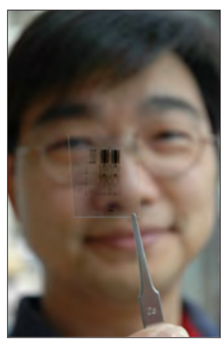
Meanwhile, technology enthusiasts continually shake their head. We all know that the true power of a picture lies in the image sensor. The battle between the two default technologies – charge-coupled device (CCD) and complementary metal oxide semiconductor (CMOS) has raged on since they were invented about 40–50 years ago.

Two camps have developed, depending on manufacturer alliances, which promote one of the two technologies.

The CCD camp touts its better picture quality, given that it has traditionally served as the benchmark. CMOS has slowly been catching up in this respect, but in all honesty, still has miles to go before it can click.

The CMOS camp, though, stresses on the smaller size of the sensors and lower power consumption. The latter is a key feature in favour of the technology, as this greatly advances its chances of becoming the *de facto* sensor type for mobile phones, where battery is king.

Size, however, is still up for debate. Yes, CMOS sensors are significantly physically smaller than CCD sensors; but they also require accompanying chips to optimise the captured colours, and of course, better quality lenses to achieve the same image quality. CCD sensors, on the other hand, bring the kitchen sink with them.



UW-Madison electrical and computer engineering graduate student Hao-Chih Yuan holds a sample of a semiconductor film on plastic

Where the companies stand...

Of course, for the end user, what matters most are for which sides the big companies are rooting.

The CMOS camp is currently championed by two heavyweights in the photo industry, Sony and Canon.

The latter has already outfitted most of its digital SLR (DSLR) cameras with CMOS sensors, replacing the once-popular CCD, and rumours are that it could soon go the same way with pro-level digital camcorders. And at this year's IEEE International Solid State Circuits Conference, the company discussed its efforts in developing a 3.3-MP CMOS image sensor that promises higher-quality video and imaging for mobile phones.

Sony, meanwhile, is concentrating on developing new techniques for its proprietary CMOS sensor's analogue-to-digital conversion process. By enabling each pixel column on the sensor to have its own conversion process, the image quality could be pushed up significantly while loss is minimised.

When it comes to mobile phones the CMOS market is booming the most, with CCD nowhere in sight – Samsung's OmniaHD is churning out 720p videos without much effort on its processor or battery.

On the CCD side, Fujitsu Research's efforts towards the Super CCD EXR have had CMOS proponents cowering. Incorporating a revised control circuit and new arrangement for colour filters, new cameras like the FinePix F200EXR offer more bang for buck than ever before.

But the greatest advocate of the CCD technology over the years has been Kodak Eastman. Last year, the company wowed the world with the world's first 50-megapixel sensor. To get an idea of the sheer scope, consider that in an aerial photo of a 2.5-kilometre-wide field, you should be able to pick out a small 1x1 foot laptop. As always, with superior photo imaging, Hasselblad is the first to jump onto the bandwagon, touting this sensor in its H3DII-50 dSLR.

Of the major companies, Nikon is the only one still straddling the fence. Some of its cameras incorporate both CMOS and CCD sensors.

And as always, the scientists who actually conduct the research on these technologies are least bothered about these cosmetic battles. For them, it's all about getting out a better picture in the end.

Clear logic

The most path-breaking technological advance that we will come to see in the near future, we think, will be from the veterans of photography – Kodak Eastman. The company has arguably done more for the world of image capture than any other. The latest barrier it has broken down brings a wide scope, as it applies to both CMOS as well as CCD sensors, and promises to end the need for flash bulbs!

In a nutshell, Kodak's researchers have figured out a way to enhance picture quality, especially at night or in motion, by two to four times that of a camera today. And they did it by adding nothing more than a 'clear' pixel to the existing Bayer pattern of filter arrays.

The improvement – by scientists John Compton and John Hamilton – could give image sensors a 2–4 times increase in sensitivity.

The two Johns have introduced a fourth pixel filter to the Bayer Pattern, which has no pigment on top. This clear pixel, called a 'panchromatic pixel', is sensitive to all visible light

Live MOS and let live

To put it simply, Live MOS seeks to take the best properties of CCD and CMOS sensors to develop a 'super-sensor'. It is claimed to deliver the image quality of a CCD, while using the kind of power of a CMOS sensor.

Essentially, it's a souped-up CMOS sensor. The biggest breakthrough for Panasonic researchers was reducing circuit paths on the surface of a CMOS sensor. By bringing these paths to the minimal two that a CCD requires, the photosensitive area of the sensor was boosted by thirty per cent.

This added clarity is what gives Panasonic the bragging rights for 'image quality that's as good as CCD', although quite a few reviewers on the internet would beg to differ.

Still, it is significantly better than CMOS in terms of picture quality, and the simplified circuit takes a lesser toll on the processor, thus providing photographs faster. Couple it with proprietary noise-reduction technology and LiveView functionality, the fast shutter speed in the compact Micro Four Thirds system promises to offer great bang for your buck.

And already, the Lumix DMC-L1 as well as the Olympus E-330 have received some accolades from the industry as well as independent reviewers. Has Panasonic really created a game-changing technology here?